PROBLEM 09

# Artificial Gravity in a Rotating Cylinder

Original statement · Problems.pdf, p. 8

Centrifugal + Coriolis Forces

In "2001: A Space Odyssey" an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

**(a)** If the cylinder has radius $R$, determine the angular velocity $\omega$ required to obtain a given level artificial gravity $g$. Consider $R = 10$ m and $g = 9.81$ m/s$^2$ for a particular case.

**(b)** An astronaut runs inside the surface of the cylinder with a constant speed $u$ with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut $h$ with respect to the radius of the cylinder $R$, $h \ll R$).

**(c)** What is the condition for her to feel no artificial gravity at all?

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Constraints, Degrees of Freedom & Coordinates

- **Physical system:** a cylindrical hull of radius $R$ spins at constant rate $\omega$ about its axis in deep space. An astronaut $P$ of mass $m$ stands on, or runs along, the inner surface.
- **Why the hull frame.** The astronaut lives in the hull: the floor, her running speed $u$ and what she "feels" are all defined with respect to the cylinder. In the frame $S_1$ attached to the hull the floor is fixed, and the only price is to add the inertia forces of Notes Eq. 4.62. The

centrifugal force is exactly the "artificial gravity" of the statement; the Coriolis force is what changes when she runs.

- **Hypotheses:**
  1. No external gravity (deep space); the only actual force on the astronaut is the total force $\mathbf{N}$ of the floor on her feet (normal reaction plus any friction; part (b) shows that, for uniform running, $\mathbf{N}$ comes out purely radial).
  2. $h \ll R$: she is a point particle on the surface, at distance $R$ from the axis.
  3. $\omega$ constant; the axis point $O$ is fixed in an inertial frame $S_0$.
- **Artificial gravity felt.** A body feels as "weight" the force it has to receive from the floor. We define the artificial gravity as $g_{\text{art}} = |\mathbf{N}|/m$, the reading per unit mass of a scale under her feet, directed opposite to $\mathbf{N}$ (i.e. away from the axis, "down" for her).
- **Degrees of freedom:** on the cylinder surface, 2 coordinates: azimuth $\varphi$ in $S_1$ and axial position $z$. Running at constant speed $u$ means $u_\theta = R\dot{\varphi}$ and $u_z = \dot{z}$ constant with $u_\theta^2 + u_z^2 = u^2$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $R$ | radius of the cylinder (10 m in the numerical case) | m |
| $\omega$ | spin rate of the cylinder | rad/s |
| $g$ | target artificial gravity (9.81 m/s$^2$) | m/s$^2$ |
| $u$ | running speed relative to the cylinder | m/s |
| $u_\theta$, $u_z$ | circumferential (positive in the sense of rotation) and axial components of the running velocity | m/s |
| $m$ | mass of the astronaut | kg |

PHASE 2

### Reference Frames & Vector Bases

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with $O$ on the axis and $\mathbf{k}_0$ along it; $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the hull, $\mathbf{k}_1 = \mathbf{k}_0$. The origin of $S_1$ is the fixed point $O$.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

(the columns are unit vectors and mutually orthogonal).

- **Motion of $S_1$: $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$** ($\omega$ constant and $\mathbf{k}_1$ fixed), $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Local cylindrical basis of the astronaut (in $S_1$).** With $\varphi$ her azimuth measured in $S_1$: $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1$ (from the axis to the floor), $\mathbf{e}_\theta = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$ (sense of rotation), $\mathbf{k}_1$. The matrix with these columns is a simple rotation by $\varphi$ (matrix $\begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\varphi + \sin^2\varphi = 1$; $RR^T$ has diagonal entries $\cos^2\varphi + \sin^2\varphi = 1$ (and 1) and off-diagonal entries $\cos\varphi\sin\varphi - \sin\varphi\cos\varphi = 0$, so $RR^T = I$). Derivatives in $S_1$ (chain rule): $\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = \dot\varphi\,\mathbf{e}_\theta$, $\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = -\dot\varphi\,\mathbf{e}_r$.
- **Inertia forces** (Notes Eq. 4.62): $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = \mathbf{0}$; $\mathbf{F}_C = m\omega^2\mathbf{r}_\perp$ (Slide 30); $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Slide 31).

PHASE 3

## Derivation

### 3.1 Kinematics

In part (a) the astronaut is at rest on the floor, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In part (b) she runs with a constant relative velocity:

**Relative kinematics.** $\mathbf{r}_1^P = R\,\mathbf{e}_r + z\,\mathbf{k}_1$ with $R\dot\varphi = u_\theta$, $\dot z = u_z$ (both constant). Differentiating in $S_1$ with the product and chain rules:

$$\mathbf{v}_1^P = R\dot\varphi\,\mathbf{e}_\theta + \dot z\,\mathbf{k}_1 = u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1$$

$$\mathbf{a}_1^P = \dot u_\theta\,\mathbf{e}_\theta + u_\theta\,(-\dot\varphi\,\mathbf{e}_r) + \dot u_z\,\mathbf{k}_1 = -\frac{u_\theta^2}{R}\,\mathbf{e}_r$$

(the relative path is a helix of radius $R$; in $S_1$ she needs a centripetal acceleration $u_\theta^2/R$).

### 3.2 Forces

The only actual force on the astronaut is the floor force $\mathbf{N}$ (no gravity in deep space). The inertia forces of Notes Eq. 4.62 are $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$ and, for the running case:

**Inertia forces.** $\mathbf{r}_\perp = R\,\mathbf{e}_r$ (the part of $\mathbf{r}_1^P$ normal to $\boldsymbol{\omega}_{10}$; the cross products are written out in part (a) below), so $\mathbf{F}_C = m\omega^2 R\,\mathbf{e}_r$. For the Coriolis force, $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$ and $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1 \times (u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1) = 2m\omega u_\theta\,\mathbf{e}_r$$

### 3.3 Equations of Motion

The unknown is the floor force $\mathbf{N}$ (the relative motion is prescribed, so no coordinate is unknown):

**Newton's law in $S_1$** (Notes Eq. 4.63), $m\mathbf{a}_1^P = \mathbf{N} + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$-m\frac{u_\theta^2}{R}\,\mathbf{e}_r = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r + 2m\omega u_\theta\,\mathbf{e}_r$$

$$\mathbf{N} = -m\left(\omega^2 R + 2\omega u_\theta + \frac{u_\theta^2}{R}\right)\mathbf{e}_r = -\frac{m}{R}\left(\omega R + u_\theta\right)^2\mathbf{e}_r$$

### 3.4 Integration and Answers

**(a) Spin rate for a given artificial gravity**

> **Pedagogical Justification:** an astronaut at rest on the floor has $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$, so Notes Eq. 4.63 becomes a static balance in $S_1$ between the floor reaction and the centrifugal force (the Coriolis force vanishes because $\mathbf{v}_1^P = \mathbf{0}$).

Position $\mathbf{r}_1^P = R\,\mathbf{e}_r$ (plus an axial part along $\mathbf{k}_1$ that does not contribute to $\mathbf{r}_\perp$), so $\mathbf{r}_\perp = R\,\mathbf{e}_r$ and

$$\mathbf{F}_C = -m\,\boldsymbol{\omega}\mathbf{k}_1 \times (\boldsymbol{\omega}\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\boldsymbol{\omega}\mathbf{k}_1 \times (R\omega\,\mathbf{e}_\theta) = m\omega^2 R\,\mathbf{e}_r$$

using $\mathbf{k}_1 \times \mathbf{e}_r = \mathbf{e}_\theta$ and $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$. Then

$$m\,\mathbf{0} = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r \quad\Longrightarrow\quad \mathbf{N} = -m\omega^2 R\,\mathbf{e}_r, \qquad g_{\text{art}} = \frac{|\mathbf{N}|}{m} = \omega^2 R$$

The floor pushes her toward the axis, and she feels a "gravity" $\omega^2 R$ pointing away from the axis. Setting $g_{\text{art}} = g$:

$$\boxed{\omega = \sqrt{\frac{g}{R}}}$$

**Numbers** ($R = 10$ m, $g = 9.81$ m/s$^2$):

$$\omega = \sqrt{\frac{9.81\ \text{m/s}^2}{10\ \text{m}}} = \sqrt{0.981}\ \text{s}^{-1} \approx 0.9905\ \text{rad/s}, \qquad T_{\text{rev}} = \frac{2\pi}{\omega} \approx 6.34\ \text{s}, \qquad \frac{60\,\omega}{2\pi} \approx 9.46\ \text{rpm}$$

and the floor moves at $\omega R \approx 9.90$ m/s in $S_0$.

**(b) Artificial gravity felt while running**

> **Pedagogical Justification:** now $\mathbf{v}_1^P \neq \mathbf{0}$, so the Coriolis force appears, and her relative path may be curved, so $\mathbf{a}_1^P \neq \mathbf{0}$. Both are computed in $S_1$ and inserted in Notes Eq. 4.63 to obtain $\mathbf{N}$. The statement does not fix the running direction, so we take a general constant relative velocity with circumferential part $u_\theta$ and axial part $u_z$ (running "around" the hull is $u_z = 0$, $u_\theta = \pm u$).

The reaction is purely radial, so no friction is needed to keep the motion uniform. The artificial gravity she feels is

$$\boxed{g_{\text{art}} = \frac{(\omega R + u_\theta)^2}{R} = g\left(1 + \frac{u_\theta}{\omega R}\right)^2 = g + 2\omega u_\theta + \frac{u_\theta^2}{R}}$$

using $\omega^2 R = g$ from part (a). Particular cases:

- running around the hull **in the sense of rotation** ($u_\theta = +u$): $g_{\text{art}} = (\omega R + u)^2/R > g$ (she feels heavier);
- running **against the rotation** ($u_\theta = -u$): $g_{\text{art}} = (\omega R - u)^2/R < g$ for $u < 2\omega R$ (she feels lighter);
- running **along the axis** ($u_\theta = 0$, $u_z = u$): $g_{\text{art}} = g$ (the Coriolis force vanishes because $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).

**Numbers** ($R = 10$ m, $\omega = 0.9905$ rad/s): at $u = 2$ m/s, $g_{\text{art}} = 14.17$ m/s$^2$ ($1.44\,g$) with the rotation and $6.25$ m/s$^2$ ($0.64\,g$) against it; at $u = 5$ m/s, $22.2$ m/s$^2$ ($2.26\,g$) and $2.41$ m/s$^2$ ($0.25\,g$).

**(c) Condition for no artificial gravity**

> **Pedagogical Justification:** $g_{\text{art}} = (\omega R + u_\theta)^2/R$ is a perfect square, so it vanishes only when its base vanishes.

$$g_{\text{art}} = 0 \iff \omega R + u_\theta = 0 \iff \boxed{u_\theta = -\omega R, \quad u_z = 0}$$

She must run around the hull, against the rotation, at the speed of the floor $\omega R$ ($\approx 9.90$ m/s for the numerical case, a sprinter's speed). Then, by Notes Eq. 3.26, her absolute velocity is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\omega R\,\mathbf{e}_\theta + \omega R\,\mathbf{e}_\theta = \mathbf{0}$: she is at rest in inertial space, floating, while the floor slides under her feet. With an axial component ($u_z \neq 0$) and $u = \omega R$ the condition cannot be met, since then $|u_\theta| < \omega R$.

PHASE 4

### Interpretation & Checks

- **Inertial reading (cross-check).** In $S_0$ the astronaut moves on a circle (or helix) of radius $R$ with tangential speed $\omega R + u_\theta$, so the floor must supply the centripetal force $m(\omega R + u_\theta)^2/R$: the same result. The rotating observer splits it into centrifugal $m\omega^2 R$, Coriolis $2m\omega u_\theta$ and relative centripetal $mu_\theta^2/R$ parts.
- **Limits:** $u_\theta \to 0$ gives $g_{\text{art}} = g$; for $|u_\theta| \ll \omega R$, $g_{\text{art}} \approx g \pm 2\omega u$, a correction due to the Coriolis force alone; a larger hull ($R \to \infty$ at fixed $g$) needs a smaller $\omega$ and makes the Coriolis correction $2\omega u = 2u\sqrt{g/R}$ smaller, which is why large habitats are more comfortable.
- **Dimensions:** $[\sqrt{g/R}] = \sqrt{\mathrm{m\,s^{-2}/m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = [2\omega u_\theta] = [u_\theta^2/R] = \mathrm{m\,s^{-2}}$; $[N] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$.
- **Verification (sympy):** $\mathbf{a}_1^P$, $\mathbf{F}_C$, $\mathbf{F}_{COR}$ and $\mathbf{N}$ are recomputed; the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition and gives the same $\mathbf{N}$; the numbers above are reproduced.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 9 in its solution keys, so the results are verified by independent derivation (inertial reading and sympy check of Phase 4).

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*Source: Problems.pdf (PDF page 8); no official key*